Online Hospital Appointment Booking System

AIM

To design and develop an Online Hospital Appointment Booking System using HTML, CSS, and JavaScript that allows patients to register, log in, search for doctors by specialty, view available appointment slots, book or cancel appointments, view appointment history, and update personal details. The system also provides an admin panel to manage doctors and appointments.

CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>MedCare Hospital Appointment System</title>

<style>
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #eef7fb;
  color: #333;
}

header {
  background: #087e8b;
  color: white;
  text-align: center;
  padding: 22px;
}

nav {
```

```
  background: #07545b;
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
}

nav button {
  background: none;
  color: white;
  border: none;
  padding: 13px;
  cursor: pointer;
}

nav button:hover {
  background: #0995a3;
}

.container {
  max-width: 1000px;
  margin: 25px auto;
  padding: 15px;
}

.page {
  display: none;
  background: white;
  padding: 24px;
  border-radius: 10px;
  box-shadow: 0 3px 10px #c9dfe6;
}

.page.active {
  display: block;
```

```
}

h2 {
  color: #087e8b;
}

input, select, textarea {
  width: 100%;
  padding: 11px;
  margin: 7px 0 15px;
  border: 1px solid #ccc;
  border-radius: 5px;
}

.action {
  background: #087e8b;
  color: white;
  border: none;
  padding: 11px 16px;
  border-radius: 5px;
  cursor: pointer;
  margin: 4px;
}

.action:hover {
  background: #07545b;
}

.doctors {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 16px;
  margin-top: 20px;
}
```

```
.card {
  border: 1px solid #cde3e9;
  padding: 16px;
  border-radius: 8px;
  background: #f9fdff;
  margin: 10px 0;
}

.card h3 {
  color: #087e8b;
}

.message {
  color: #16834a;
  font-weight: bold;
}

footer {
  background: #07545b;
  color: white;
  text-align: center;
  padding: 15px;
  margin-top: 30px;
}

@media (max-width: 600px) {
  .container {
    margin: 10px;
    padding: 8px;
  }

  nav button {
    padding: 10px;
```

```
  }
}
</style>
</head>

<body>

<header>
  <h1>MedCare Hospital</h1>
  <p>Online Doctor Appointment Booking System</p>
</header>

<nav>
  <button onclick="showPage('home')">Home</button>
  <button onclick="showPage('register')">Register</button>
  <button onclick="showPage('login')">Login</button>
  <button onclick="showPage('doctors')">Find Doctors</button>
  <button onclick="showPage('booking')">Book Appointment</button>
  <button onclick="showPage('appointments')">My Appointments</button>
  <button onclick="showPage('profile')">My Profile</button>
  <button onclick="showPage('admin')">Admin Panel</button>
</nav>

<div class="container">

<!-- Home Page -->
<section id="home" class="page active">
  <h2>Welcome to MedCare Hospital</h2>
  <p>Book appointments with experienced doctors from different specialties.</p>
  <p>Choose your doctor, select a convenient date and time, and manage your appointments online.</p>
  <button class="action" onclick="showPage('doctors')">Find a Doctor</button>
  <button class="action" onclick="showPage('booking')">Book Appointment</button>
</section>
```

```
<!-- Registration -->
<section id="register" class="page">
  <h2>Patient Registration</h2>

  <label>Full Name</label>
  <input id="regName" type="text" placeholder="Enter your full name">

  <label>Email</label>
  <input id="regEmail" type="email" placeholder="Enter your email">

  <label>Phone Number</label>
  <input id="regPhone" type="tel" placeholder="Enter your phone number">

  <label>Password</label>
  <input id="regPassword" type="password" placeholder="Create a password">

  <button class="action" onclick="registerPatient()">Register</button>
  <p id="registerMessage" class="message"></p>
</section>

<!-- Login -->
<section id="login" class="page">
  <h2>Patient Login</h2>

  <label>Email</label>
  <input id="loginEmail" type="email" placeholder="Enter your email">

  <label>Password</label>
  <input id="loginPassword" type="password" placeholder="Enter your password">

  <button class="action" onclick="loginPatient()">Login</button>
  <p id="loginMessage" class="message"></p>
</section>
```

```
<!-- Find Doctors -->
<section id="doctors" class="page">
  <h2>Find a Doctor</h2>

  <label>Search by Specialty</label>
  <select id="specialtyFilter" onchange="displayDoctors()">
    <option value="All">All Specialties</option>
    <option>Cardiology</option>
    <option>Dermatology</option>
    <option>Orthopedics</option>
    <option>Pediatrics</option>
    <option>General Medicine</option>
    <option>Neurology</option>
  </select>

  <label>Search Hospital or Location</label>
  <input id="hospitalSearch" type="text"
      placeholder="Enter hospital or city"
      oninput="displayDoctors()">

  <div id="doctorList" class="doctors"></div>
</section>

<!-- Booking -->
<section id="booking" class="page">
  <h2>Book an Appointment</h2>

  <label>Select Doctor</label>
  <select id="doctorChoice" onchange="showSlots()"></select>

  <label>Appointment Date</label>
  <input type="date" id="appointmentDate" onchange="showSlots()">

  <label>Available Time Slot</label>
```

```
    <select id="appointmentTime">
      <option value="">Select a date first</option>
    </select>

    <label>Reason for Visit</label>
    <textarea id="reason" rows="3"
          placeholder="Briefly describe the reason for your visit"></textarea>

    <button class="action" onclick="bookAppointment()">Confirm Appointment</button>
    <p id="bookingMessage" class="message"></p>
  </section>

  <!-- My Appointments -->
  <section id="appointments" class="page">
    <h2>My Appointments and History</h2>
    <button class="action" onclick="displayAppointments()">Refresh</button>
    <div id="appointmentList"></div>
  </section>

  <!-- Patient Profile -->
  <section id="profile" class="page">
    <h2>My Personal Details</h2>

    <label>Full Name</label>
    <input id="profileName" type="text" placeholder="Enter your name">

    <label>Email</label>
    <input id="profileEmail" type="email" disabled>

    <label>Phone Number</label>
    <input id="profilePhone" type="tel" placeholder="Enter your phone number">

    <button class="action" onclick="updateProfile()">Update Profile</button>
    <p id="profileMessage" class="message"></p>
```

```
</section>

<!-- Admin Panel -->
<section id="admin" class="page">
  <h2>Administrator Login</h2>

  <label>Username</label>
  <input id="adminUser" type="text" placeholder="Enter admin username">

  <label>Password</label>
  <input id="adminPassword" type="password" placeholder="Enter admin password">

  <button class="action" onclick="adminLogin()">Login</button>
  <p id="adminMessage" class="message"></p>

  <div id="adminPanel" style="display:none;">
    <h2>Add Doctor</h2>

    <label>Doctor Name</label>
    <input id="newDoctorName" type="text" placeholder="Enter doctor name">

    <label>Specialty</label>
    <select id="newDoctorSpecialty">
      <option>Cardiology</option>
      <option>Dermatology</option>
      <option>Orthopedics</option>
      <option>Pediatrics</option>
      <option>General Medicine</option>
      <option>Neurology</option>
    </select>

    <label>Hospital Name</label>
    <input id="newHospital" type="text" placeholder="Enter hospital name">
```

```
      <label>Consultation Fee (INR)</label>
      <input id="newFee" type="number" min="0" placeholder="Enter fee">

      <button class="action" onclick="addDoctor()">Add Doctor</button>

      <h2>Manage Appointments</h2>
      <div id="adminAppointments"></div>
    </div>
  </section>

  </div>

  <footer>
    <p>MedCare Hospital | Internet Programming Lab Experiment</p>
    <p>For emergencies, contact your local emergency service.</p>
  </footer>

  <script>
  let patients = [];
  let currentPatient = null;
  let isAdmin = false;
  let appointments = [];
  let nextAppointmentId = 1001;

  let doctors = [
    {
      id: 1,
      name: "Dr. Ravi Kumar",
      specialty: "Cardiology",
      hospital: "MedCare Main Hospital",
      fee: 700
    },
    {
      id: 2,
```

```
    name: "Dr. Priya Sharma",
    specialty: "Dermatology",
    hospital: "MedCare Main Hospital",
    fee: 500
  },
  {
    id: 3,
    name: "Dr. Arun Reddy",
    specialty: "Orthopedics",
    hospital: "City Care Hospital",
    fee: 600
  },
  {
    id: 4,
    name: "Dr. Meena Rao",
    specialty: "Pediatrics",
    hospital: "MedCare Children's Hospital",
    fee: 450
  },
  {
    id: 5,
    name: "Dr. Suresh Kumar",
    specialty: "General Medicine",
    hospital: "City Care Hospital",
    fee: 400
  },
  {
    id: 6,
    name: "Dr. Anjali Das",
    specialty: "Neurology",
    hospital: "MedCare Main Hospital",
    fee: 800
  }
];
```

```
function showPage(id) {
  document.querySelectorAll(".page").forEach(page => {
    page.classList.remove("active");
  });

  document.getElementById(id).classList.add("active");

  if (id === "doctors") displayDoctors();
  if (id === "booking") updateDoctorOptions();
  if (id === "appointments") displayAppointments();
  if (id === "profile") loadProfile();
  if (id === "admin" && isAdmin) displayAdminAppointments();
}

function registerPatient() {
  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim();
  const phone = document.getElementById("regPhone").value.trim();
  const password = document.getElementById("regPassword").value;

  if (!name || !email || !phone || !password) {
    alert("Please fill in all fields.");
    return;
  }

  if (patients.some(patient => patient.email === email)) {
    alert("This email is already registered.");
    return;
  }

  patients.push({ name, email, phone, password });

  document.getElementById("registerMessage").textContent =
```

```
    "Registration successful. Please log in.";

  document.getElementById("regName").value = "";
  document.getElementById("regEmail").value = "";
  document.getElementById("regPhone").value = "";
  document.getElementById("regPassword").value = "";
}

function loginPatient() {
  const email = document.getElementById("loginEmail").value.trim();
  const password = document.getElementById("loginPassword").value;

  const patient = patients.find(item =>
    item.email === email && item.password === password
  );

  if (!patient) {
    document.getElementById("loginMessage").textContent =
      "Invalid login details. Please register or check your credentials.";
    return;
  }

  currentPatient = patient;

  document.getElementById("loginMessage").textContent =
    "Welcome, " + patient.name + "! Login successful.";

  showPage("profile");
}

function displayDoctors() {
  const specialty = document.getElementById("specialtyFilter").value;
  const search = document.getElementById("hospitalSearch").value.toLowerCase();
```

```
  const filtered = doctors.filter(doctor =>
    (specialty === "All" || doctor.specialty === specialty) &&
    (doctor.name.toLowerCase().includes(search) ||
     doctor.hospital.toLowerCase().includes(search) ||
     doctor.specialty.toLowerCase().includes(search))
  );

  const list = document.getElementById("doctorList");

  if (filtered.length === 0) {
    list.innerHTML = "<p>No matching doctors found.</p>";
    return;
  }

  list.innerHTML = filtered.map(doctor => `
    <div class="card">
      <h3>${doctor.name}</h3>
      <p>Specialty: ${doctor.specialty}</p>
      <p>Hospital: ${doctor.hospital}</p>
      <p>Consultation Fee: ₹${doctor.fee}</p>
      <button class="action" onclick="selectDoctor(${doctor.id})">
        Book Appointment
      </button>
    </div>
  `).join("");
}

function updateDoctorOptions() {
  const select = document.getElementById("doctorChoice");

  select.innerHTML = doctors.map(doctor =>
    `<option value="${doctor.id}">${doctor.name} - ${doctor.specialty}</option>`
  ).join("");
```

```
  showSlots();
}

function selectDoctor(id) {
  showPage("booking");
  document.getElementById("doctorChoice").value = String(id);
  showSlots();
}

function showSlots() {
  const doctorId = Number(document.getElementById("doctorChoice").value);
  const date = document.getElementById("appointmentDate").value;
  const select = document.getElementById("appointmentTime");

  const times = [
    "09:00", "10:00", "11:00",
    "12:00", "14:00", "15:00", "16:00"
  ];

  const today = new Date();
  today.setHours(0, 0, 0, 0);

  if (!date) {
    select.innerHTML = '<option value="">Select a date first</option>';
    return;
  }

  if (new Date(date + "T00:00:00") < today) {
    select.innerHTML = '<option value="">Choose a future date</option>';
    return;
  }

  const available = times.filter(time =>
    !appointments.some(item =>
```

```
      item.doctorId === doctorId &&
      item.date === date &&
      item.time === time &&
      item.status !== "Cancelled"
    )
  );

  select.innerHTML = available.length
    ? available.map(time =>
      `<option value="${time}">${time}</option>`
    ).join("")
    : '<option value="">No available slots</option>';
}

function bookAppointment() {
  if (!currentPatient) {
    alert("Please register and log in first.");
    showPage("register");
    return;
  }

  const doctorId = Number(document.getElementById("doctorChoice").value);
  const doctor = doctors.find(item => item.id === doctorId);
  const date = document.getElementById("appointmentDate").value;
  const time = document.getElementById("appointmentTime").value;
  const reason = document.getElementById("reason").value.trim();

  if (!doctor || !date || !time) {
    alert("Please select a doctor, date, and available time.");
    return;
  }

  const slotTaken = appointments.some(item =>
    item.doctorId === doctorId &&
```

```
    item.date === date &&
    item.time === time &&
    item.status !== "Cancelled"
  );

  if (slotTaken) {
    alert("This appointment slot is no longer available. Please choose another.");
    showSlots();
    return;
  }

  const appointment = {
    id: "MC" + nextAppointmentId++,
    patientName: currentPatient.name,
    email: currentPatient.email,
    phone: currentPatient.phone,
    doctorId: doctor.id,
    doctorName: doctor.name,
    specialty: doctor.specialty,
    hospital: doctor.hospital,
    fee: doctor.fee,
    date,
    time,
    reason,
    status: "Booked"
  };

  appointments.push(appointment);

  document.getElementById("bookingMessage").textContent =
    "Appointment booked successfully! Your appointment ID is " +
    appointment.id + ".";

  document.getElementById("reason").value = "";
```

```
  displayAppointments();
  displayAdminAppointments();
  showSlots();
}

function displayAppointments() {
  const list = document.getElementById("appointmentList");

  if (!currentPatient) {
    list.innerHTML = "<p>Please log in to view your appointments.</p>";
    return;
  }

  const mine = appointments.filter(item =>
    item.email === currentPatient.email
  );

  if (mine.length === 0) {
    list.innerHTML = "<p>You have no appointments yet.</p>";
    return;
  }

  list.innerHTML = mine.map(item => `
    <div class="card">
      <h3>${item.doctorName}</h3>
      <p>Appointment ID: ${item.id}</p>
      <p>Specialty: ${item.specialty}</p>
      <p>Hospital: ${item.hospital}</p>
      <p>Date: ${item.date}</p>
      <p>Time: ${item.time}</p>
      <p>Consultation Fee: ₹${item.fee}</p>
      <p>Status: <strong>${item.status}</strong></p>
      ${item.status !== "Cancelled" ? `
```

```
            <button class="action" onclick="cancelAppointment('${item.id}')">
              Cancel Appointment
            </button>
          ` : ""}
        </div>
      `).join("");
}

function cancelAppointment(id) {
  const appointment = appointments.find(item => item.id === id);

  if (!appointment || appointment.status === "Cancelled") return;

  appointment.status = "Cancelled";

  alert("Appointment cancelled successfully.");

  displayAppointments();
  showSlots();
  displayAdminAppointments();
}

function loadProfile() {
  if (!currentPatient) {
    document.getElementById("profileName").value = "";
    document.getElementById("profileEmail").value = "";
    document.getElementById("profilePhone").value = "";
    document.getElementById("profileMessage").textContent =
      "Please log in to view your profile.";
    return;
  }

  document.getElementById("profileName").value = currentPatient.name;
  document.getElementById("profileEmail").value = currentPatient.email;
```

document.getElementById("profilePhone")

